

STANDARD 2

Stage 6

Measurement (Std 2), M6 Trigonometry (Y12)

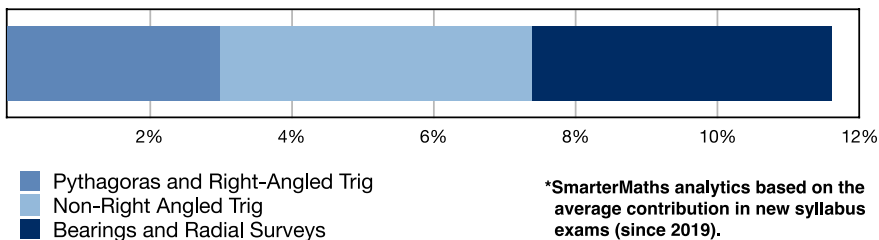
Non-Right Angled Trig (Std2)

Teacher: Hailey Summer

Exam Equivalent Time: 112.5 minutes (based on allocation of 1.5 minutes per mark)

STD2 Exam Contribution History

M6 - Trigonometry



IMPORTANT FEATURES AND TIPS FROM EXAM HISTORY

- MS-M6 Trigonometry* has been a very significant contributor to new syllabus Std2 exams, responsible for an average of 11.6% per exam since 2019.
- We have split this area into 3 subtopics: *Pythagoras and Right-Angled Trig* (3.0%), *Non-Right-Angled Trig* (4.4%) and *Bearings* (4.2%).
- This analysis looks at *Non-Right-Angled Trig* (4.4%).

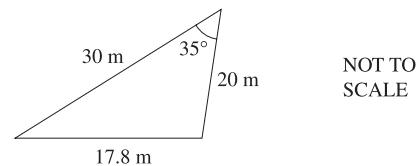
ANALYSIS - What to Expect and Common pitfalls

- The most difficult question type in this topic area is undoubtedly "2-triangle" questions which have produced sub-50% mean marks more often than not when examined in the last decade.
- "2-Triangle" questions have been asked 7 times since 2009 (most recently in 2022 after being omitted from 2019-2021) and should be a focus of any revision.
- Other more standard *Non Right-Angled Trig* questions requiring students to simply apply a version of the sine/cosine formula, have been regularly tested and generally well answered with the exception of 2021 Q37 (involving an obtuse angle) and 2010 26d which produced mean marks of 31% and 32% respectively.
- While *Non Right-Angled Trig* calculations are a core competency within *Bearings*, note this analysis and allocation history does not include *Bearings* which are dealt with in their own sub-topic.

Questions

1. Measurement, STD2 M6 2004 HSC 9 MC

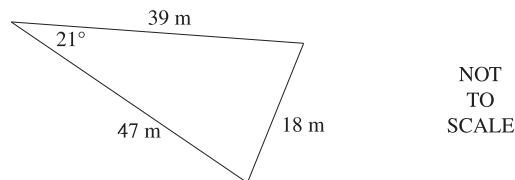
What is the area of the triangle to the nearest square metre?



- A. 102 m²
- B. 153 m²
- C. 172 m²
- D. 178 m²

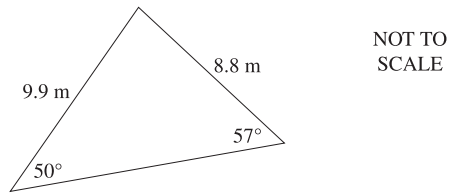
2. Measurement, STD2 M6 2006 HSC 9 MC

What is the area of this triangle, to the nearest square metre?



- A. 152 m²
- B. 283 m²
- C. 328 m²
- D. 351 m²

3. Measurement, STD2 M6 2012 HSC 10 MC



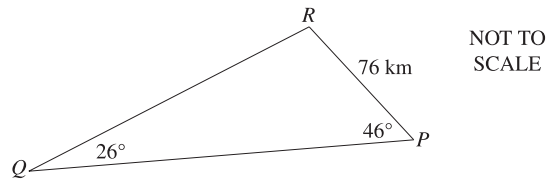
What is the area of this triangle, to the nearest square metre?

- A. 33 m²
- B. 37 m²
- C. 42 m²
- D. 44 m²

4. Measurement, STD2 M6 2010 HSC 9 MC

Three towns **P**, **Q** and **R** are marked on the diagram.

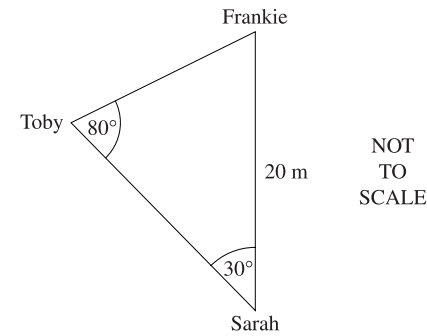
The distance from **R** to **P** is 76 km. $\angle RQP = 26^\circ$ and $\angle RPQ = 46^\circ$.



What is the distance from **P** to **Q** to the nearest kilometre?

- A. 100 km
- B. 125 km
- C. 165 km
- D. 182 km

5. Measurement, STD2 M6 2005 HSC 5 MC

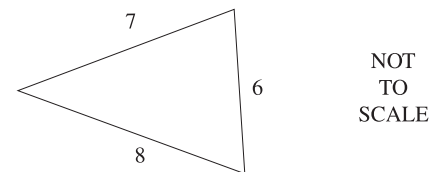


Which formula should be used to calculate the distance between Toby and Frankie?

- A. $\frac{a}{\sin A} = \frac{b}{\sin B}$
- B. $c^2 = a^2 + b^2$
- C. $A = \frac{1}{2}ab \sin C$
- D. $c^2 = a^2 + b^2 - 2ab \cos C$

6. Measurement, STD2 M6 2008 HSC 5 MC

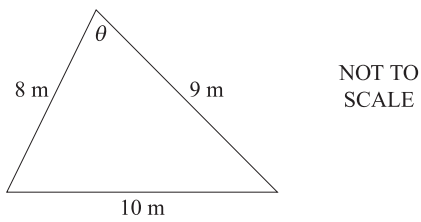
What is the size of the smallest angle in this triangle?



- A. 29°
- B. 47°
- C. 58°
- D. 76°

7. Measurement, STD2 M6 2018 HSC 12 MC

The diagram shows a triangle with side lengths 8 m, 9 m and 10m.

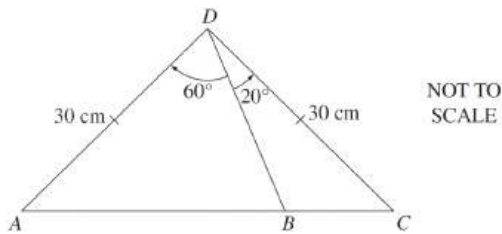


What is the value of θ , marked on the diagram, to the nearest degree?

- A. 49°
- B. 51°
- C. 59°
- D. 72°

8. Measurement, STD2 M6 2009 HSC 22 MC

In the diagram, AD and DC are equal to 30 cm.

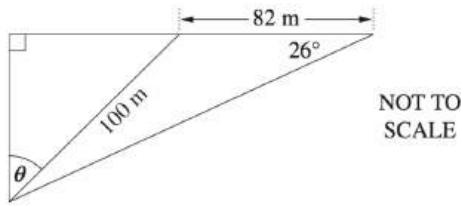


What is the length of AB to the nearest centimetre?

- A. 28 cm
- B. 31 cm
- C. 34 cm
- D. 39 cm

9. Measurement, STD2 M6 2013 HSC 24 MC

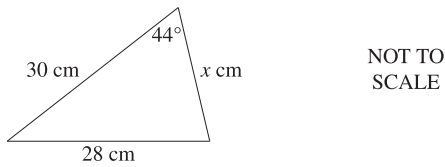
What is the value of θ , to the nearest degree?



- A. 21°
- B. 32°
- C. 43°
- D. 55°

10. Measurement, STD2 M6 2015 HSC 22 MC

The area of the triangle shown is 250 cm².

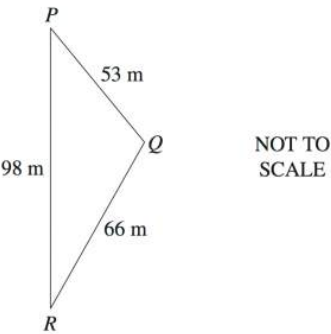


What is the value of x , correct to the nearest whole number?

- A. 11
- B. 18
- C. 22
- D. 24

11. Measurement, STD2 M6 2013 HSC 26a

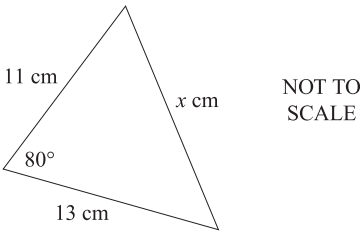
Triangle **PQR** is shown.



Find the size of angle **Q**, to the nearest degree. (2 marks)

12. Measurement, STD2 M6 2019 HSC 17

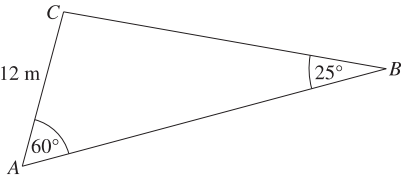
The diagram shows a triangle with sides of length **x** cm, 11 cm and 13 cm and an angle of 80°.



Use the cosine rule to calculate the value of **x**, correct to two significant figures. (3 marks)

13. Measurement, STD2 M6 2023 HSC 35

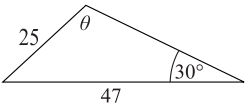
The diagram shows triangle **ABC**.



Calculate the area of the triangle, to the nearest square metre. (3 marks)

14. Measurement, STD2 M6 SM-Bank 1

The diagram shows a triangle with side lengths 25 cm and 47 cm and angle 30° and **θ**.

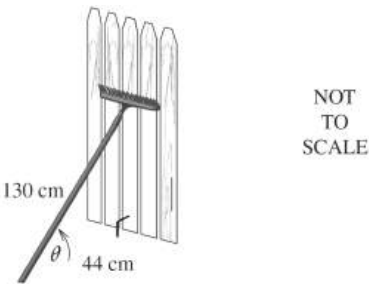


Find **θ** given it is an obtuse angle. Give your answer to the nearest minute. (3 marks)

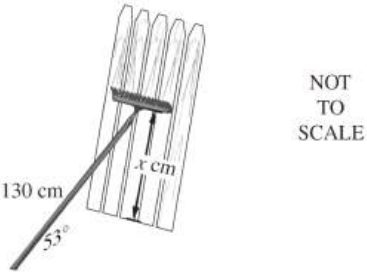
15. Measurement, STD2 M6 2006 HSC 24b

A 130 cm long garden rake leans against a fence. The end of the rake is 44 cm from the base of the fence.

i. If the fence is vertical, find the value of **θ** to the nearest degree. (2 marks)

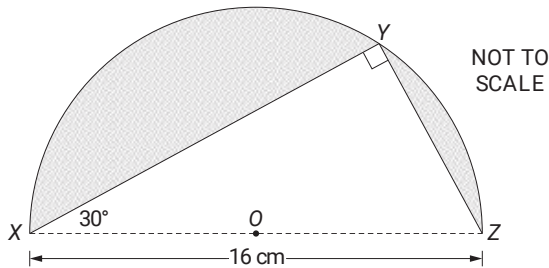


ii. The fence develops a lean and the rake is now at an angle of 53° to the ground. Calculate the new distance (**x** cm) from the base of the fence to the head of the rake. Give your answer to the nearest centimetre. (2 marks)



16. Measurement, STD2 M6 2021 HSC 32

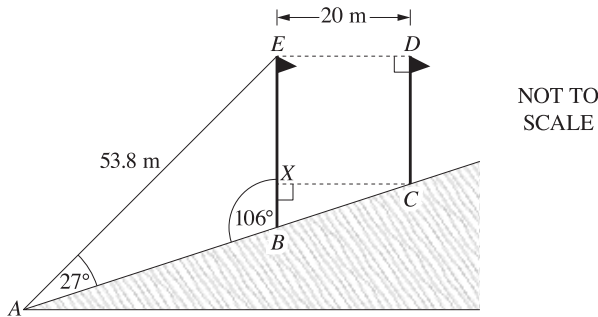
A right-angled triangle XYZ is cut out from a semicircle with centre O . The length of the diameter XZ is 16 cm and $\angle YXZ = 30^\circ$, as shown on the diagram.



- a. Find the length of XY in centimetres, correct to two decimal places. (2 marks)
- b. Hence, find the area of the shaded region in square centimetres, correct to one decimal place. (3 marks)

17. Measurement, STD2 M6 2024 HSC 36

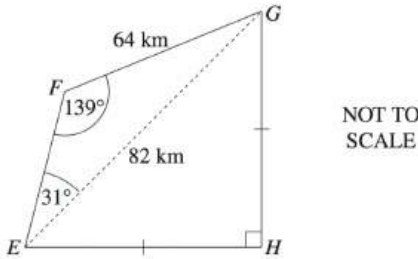
The diagram shows two vertical flagpoles, BE and CD , set on sloping ground.



- a. What is the height of the flagpole BE , correct to 1 decimal place? (2 marks)
- b. What is the height of the flagpole CD , correct to 1 decimal place? (2 marks)

18. Measurement, STD2 M6 2012 HSC 29c

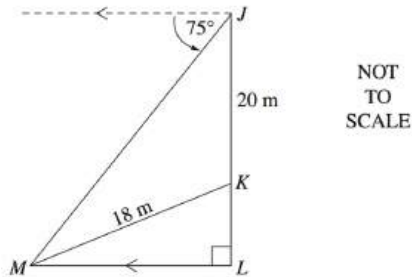
Raj cycles around a course. The course starts at E , passes through F , G and H and finishes at E . The distances EH and GH are equal.



- i. What is the length of EF , to the nearest kilometre? (2 marks)
- ii. What is the total distance that Raj cycles, to the nearest kilometre? (3 marks)

19. Measurement, STD2 M6 2007 HSC 25b

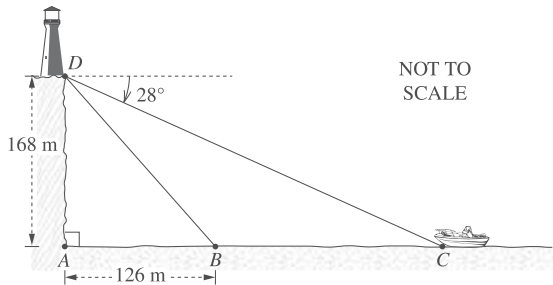
The angle of depression from J to M is 75° . The length of JK is 20 m and the length of MK is 18 m.



Copy or trace this diagram into your writing booklet and calculate the angle of elevation from M to K . Give your answer to the nearest degree. (3 marks)

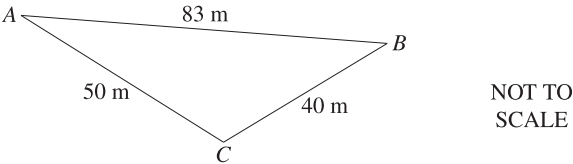
20. Measurement, STD2 M6 2010 HSC 24d

The base of a lighthouse, **D**, is at the top of a cliff 168 metres above sea level. The angle of depression from **D** to a boat at **C** is 28° . The boat heads towards the base of the cliff, **A**, and stops at **B**. The distance **AB** is 126 metres.



- i. What is the angle of depression from **D** to **B**, correct to the nearest degree? (3 marks)
- ii. How far did the boat travel from **C** to **B**, correct to the nearest metre? (2 marks)

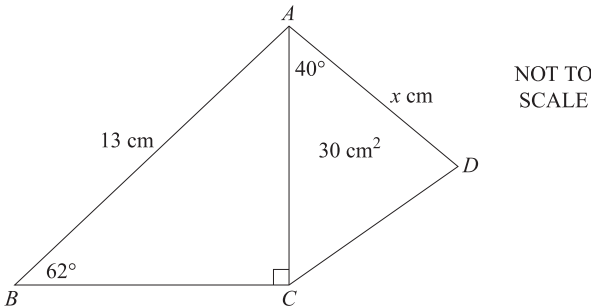
21. Measurement, STD2 M6 2010 HSC 26d



Find the area of triangle **ABC**, correct to the nearest square metre. (3 marks)

22. Measurement, STD2 M6 2018 HSC 30c

The diagram shows two triangles.
Triangle **ABC** is right-angled, with **AB** = 13cm and $\angle ABC = 62^\circ$.
In triangle **ACD**, **AD** = x cm and $\angle DAC = 40^\circ$. The area of triangle **ACD** is 30 cm².

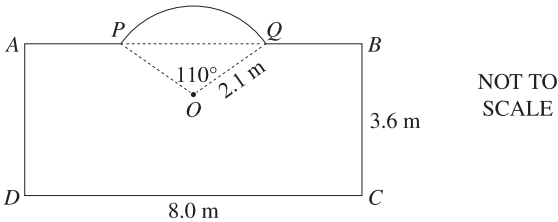


What is the value of **x**, correct to one decimal place? (3 marks)

23. Measurement, STD2 M1 2023 HSC 33

The diagram shows a shape **APQBCD**. The shape consists of a rectangle **ABCD** with an arc **PQ** on side **AB** and with side lengths **BC** = 3.6 m and **CD** = 8.0 m.

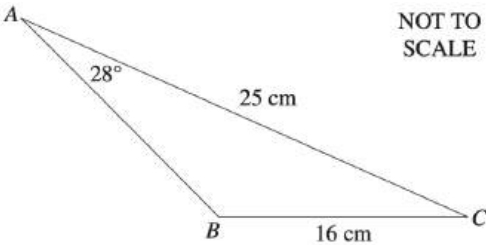
The arc **PQ** is an arc of a circle with centre **O** and radius 2.1 m and $\angle POQ = 110^\circ$.



What is the perimeter of the shape **APQBCD**? Give your answer correct to one decimal place. (4 marks)

24. Measurement, STD2 M6 2021 HSC 37

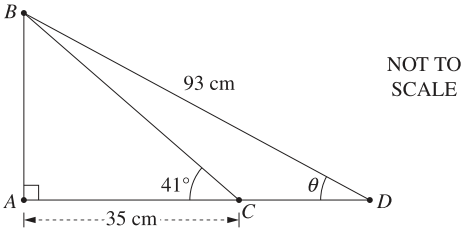
The diagram shows a triangle **ABC** where **AC** = 25 cm, **BC** = 16 cm, **∠BAC** = 28° and angle **ABC** is obtuse.



Find the size of the obtuse angle **ABC** correct to the nearest degree. (3 marks)

25. Measurement, STD2 M6 2022 HSC 26

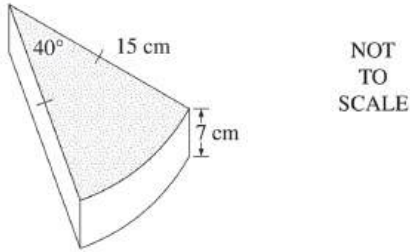
The diagram shows two right-angled triangles, **ABC** and **ABD**, where **AC** = 35 cm, **BD** = 93 cm, **∠ACB** = 41° and **∠ADB** = θ .



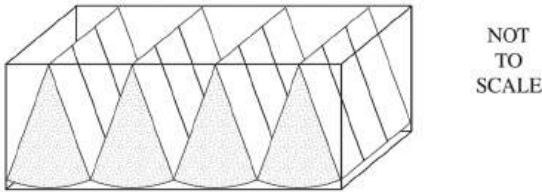
Calculate the size of angle θ , to the nearest minute. (4 marks)

26. Measurement, STD2 M1 2008 HSC 25c

Pieces of cheese are cut from cylindrical blocks with dimensions as shown.

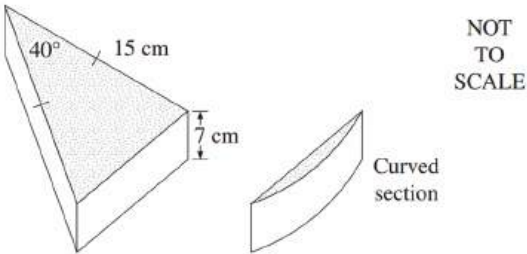


Twelve pieces are packed in a rectangular box. There are three rows with four pieces of cheese in each row. The curved surface is face down with the pieces touching as shown.



i. What are the dimensions of the rectangular box? (4 marks)

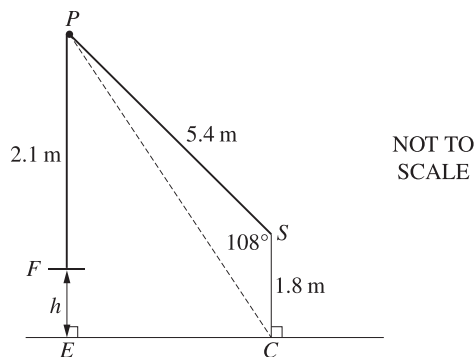
To save packing space, the curved section is removed.



ii. What is the volume of the remaining triangular prism of cheese? Answer to the nearest cubic centimetre. (2 marks)

27. Measurement, STD2 M6 2015 HSC 30e

From point S , which is 1.8 m above the ground, a pulley at P is used to lift a flat object F . The lengths SP and PF are 5.4 m and 2.1 m respectively. The angle PSC is 108° .



- Show that the length PC is 6.197 m, correct to 3 decimal places. (1 mark)
- Calculate h , the height of the object above the ground. (4 marks)

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Worked Solutions

1. Measurement, STD2 M6 2004 HSC 9 MC

Using sine rule,

$$\begin{aligned} \text{Area} &= \frac{1}{2}ab\sin C \\ &= \frac{1}{2} \times 30 \times 20 \times \sin 35^\circ \\ &= 172.072\dots \text{ m}^2 \\ &\Rightarrow C \end{aligned}$$

2. Measurement, STD2 M6 2006 HSC 9 MC

Using the Sine rule

$$\begin{aligned} A &= \frac{1}{2}ab \sin C \\ &= \frac{1}{2} \times 39 \times 47 \times \sin 21^\circ \\ &= 328.44\dots \text{ m}^2 \\ &\Rightarrow C \end{aligned}$$

3. Measurement, STD2 M6 2012 HSC 10 MC

Let unknown angle = $\angle C$

$$\begin{aligned} \angle C &= 180 - (50 + 57) \quad (180^\circ \text{ in } \Delta) \\ &= 73^\circ \end{aligned}$$

$$\begin{aligned} \therefore A &= \frac{1}{2}ab \sin C \\ &= \frac{1}{2} \times 9.9 \times 8.8 \times \sin 73^\circ \\ &= 41.656 \text{ m}^2 \\ &\Rightarrow C \end{aligned}$$

4. Measurement, STD2 M6 2010 HSC 9 MC

$$\begin{aligned}\angle QRP &= 180 - (26 + 46) \quad (180^\circ \text{ in } \Delta) \\ &= 108^\circ\end{aligned}$$

Using sine rule

$$\begin{aligned}\frac{PQ}{\sin 108^\circ} &= \frac{76}{\sin 26^\circ} \\ PQ &= \frac{76 \times \sin 108^\circ}{\sin 26^\circ} \\ &= 164.88 \text{ km} \\ \Rightarrow C\end{aligned}$$

5. Measurement, STD2 M6 2005 HSC 5 MC

The triangle is not a right-angled triangle,
 $\therefore$ Not B
Given the information on the diagram provides
2 angles and 1 side, the sine rule will work best.

$$\begin{aligned}\frac{a}{\sin A} &= \frac{b}{\sin B} \\ \Rightarrow A\end{aligned}$$

6. Measurement, STD2 M6 2008 HSC 5 MC

Smallest angle is opposite smallest side.

$$\begin{aligned}\cos A &= \frac{b^2 + c^2 - a^2}{2bc} \\ &= \frac{7^2 + 8^2 - 6^2}{2 \times 7 \times 8} \\ &= 0.6875 \\ A &= \cos^{-1}(0.6875) \\ \therefore A &= 46.567 \dots^\circ \\ \Rightarrow B\end{aligned}$$

7. Measurement, STD2 M6 2018 HSC 12 MC

Using the cosine rule:

$$\begin{aligned}\cos \theta &= \frac{8^2 + 9^2 - 10^2}{2 \times 8 \times 9} \\ &= 0.3125 \\ \therefore \theta &= \cos^{-1}(0.3125) \\ &= 71.790 \dots^\circ\end{aligned}$$

$$\Rightarrow D$$

8. Measurement, STD2 M6 2009 HSC 22 MC

$\triangle ADC$ is isosceles

$$\begin{aligned}\angle DAB &= \angle DCA = x^\circ \\ 2x + 80^\circ &= 180^\circ \quad (\text{Angle sum of } \triangle ADC) \\ 2x &= 100^\circ \\ x &= 50^\circ\end{aligned}$$

$$\begin{aligned}\angle DBA &= 180 - (50 + 60) \quad (\text{Angle sum of } \triangle ADB) \\ &= 70^\circ\end{aligned}$$

Using sine rule:

$$\begin{aligned}\frac{AB}{\sin 60} &= \frac{30}{\sin 70} \\ AB &= \frac{30 \times \sin 60}{\sin 70} \\ &= 27.648 \dots \text{ cm} \\ \Rightarrow A\end{aligned}$$

◆ Mean mark of 35%

9. Measurement, STD2 M6 2013 HSC 24 MC

$$\frac{a}{\sin A} = \frac{b}{\sin B}$$
$$\frac{82}{\sin A} = \frac{100}{\sin 26}$$
$$\sin A = \frac{82 \times \sin 26}{100}$$
$$= 0.35946 \dots$$
$$\angle A = 21^\circ \quad (\text{nearest degree})$$

Since 180° in Δ :

$$90 + 26 + (\theta + 21) = 180$$
$$\theta = 43^\circ$$

$\Rightarrow C$

10. Measurement, STD2 M6 2015 HSC 22 MC

Using $A = \frac{1}{2}ab \sin C$

$$250 = \frac{1}{2} \times 30x \sin 44^\circ$$
$$250 = 15x \sin 44^\circ$$
$$\therefore x = \frac{250}{15 \sin 44^\circ}$$
$$= 23.99 \dots \text{ m}$$

$\Rightarrow D$

11. Measurement, STD2 M6 2013 HSC 26a

Using Cosine rule

$$\cos \angle Q = \frac{a^2 + b^2 - c^2}{2ab}$$
$$= \frac{53^2 + 66^2 - 98^2}{2 \times 53 \times 66}$$
$$= -0.3486 \dots$$

$$\therefore \angle Q = 110.4034 \dots$$
$$= 110^\circ \quad (\text{nearest degree})$$

12. Measurement, STD2 M6 2019 HSC 17

$$x^2 = 11^2 + 13^2 - 2 \times 11 \times 13 \times \cos 80^\circ$$
$$= 240.336 \dots$$

$$\therefore x = 15.502 \dots$$
$$= 16 \text{ cm} \quad (2 \text{ sig. fig.})$$

13. Measurement, STD2 M6 2023 HSC 35

Using the sine rule:

$$\frac{CB}{\sin 60^\circ} = \frac{12}{\sin 25^\circ}$$
$$CB = \sin 60^\circ \times \frac{12}{\sin 25^\circ}$$
$$= 24.590 \dots$$

$$\angle ACB = 180 - (60 + 25) = 95^\circ \quad (180^\circ \text{ in } \Delta)$$

Using the sine rule (Area):

$$A = \frac{1}{2} \times AC \times CB \times \sin \angle ACB$$
$$= \frac{1}{2} \times 12 \times 24.59 \times \sin 95^\circ$$
$$= 146.98 \dots$$
$$= 147 \text{ m}^2$$

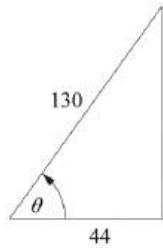
14. Measurement, STD2 M6 SM-Bank 1

Using sine rule:

$$\frac{\sin \theta}{47} = \frac{\sin 30^\circ}{25}$$
$$\sin \theta = 47 \times \frac{\frac{1}{2}}{25}$$
$$\sin \theta = \frac{47}{50}$$
$$\theta = \sin^{-1} \frac{47}{50}$$
$$= 70.05 \text{ or } 109.94$$
$$= 109.948 \dots \quad (\theta \text{ is obtuse})$$
$$= 109^\circ 57'$$

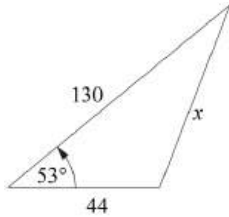
15. Measurement, STD2 M6 2006 HSC 24b

i.



$$\begin{aligned}\cos\theta &= \frac{44}{130} \\ &= 70.216\dots^\circ \\ &= 70^\circ \text{ (nearest degree)}\end{aligned}$$

ii.



Using cosine rule

$$\begin{aligned}x^2 &= 130^2 + 44^2 - 2 \times 130 \times 44 \times \cos 53^\circ \\ &= 11\,951.23\dots \\ x &= 109.32\dots \\ &= 109 \text{ cm (nearest cm)}\end{aligned}$$

16. Measurement, STD2 M6 2021 HSC 32

$$\begin{aligned}\text{a. } \cos 30^\circ &= \frac{XY}{16} \\ XY &= 16 \cos 30^\circ \\ &= 13.8564 \\ &= 13.86 \text{ cm (2 d.p.)}\end{aligned}$$

$$\begin{aligned}\text{b. Area of semi-circle} &= \frac{1}{2} \times \pi r^2 \\ &= \frac{1}{2} \pi \times 8^2 \\ &= 100.531 \text{ cm}^2\end{aligned}$$

$$\begin{aligned}\text{Area of } \triangle XYZ &= \frac{1}{2} ab \sin C \\ &= \frac{1}{2} \times 16 \times 13.856 \times \sin 30^\circ \\ &= 55.42 \text{ cm}^2\end{aligned}$$

$$\begin{aligned}\therefore \text{Shaded Area} &= 100.531 - 55.42 \\ &= 45.111 \\ &= 45.1 \text{ cm}^2 \text{ (1 d.p.)}\end{aligned}$$

♦ Mean mark part (b) 36%.

17. Measurement, STD2 M6 2024 HSC 36

a. Using the sine rule:

$$\begin{aligned}\frac{BE}{\sin 27^\circ} &= \frac{53.8}{\sin 106^\circ} \\ \therefore BE &= \frac{53.8 \times \sin 27^\circ}{\sin 106^\circ} \\ &= 25.408\dots \\ &= 25.4 \text{ m (1 d.p.)}\end{aligned}$$

b. $CD = EB - XB$

Consider $\triangle XBC$:
 $\angle XBC = 180 - 106 = 74^\circ$, $XC = ED = 20$

$$\begin{aligned}\tan 74^\circ &= \frac{20}{XB} \\ XB &= \frac{20}{\tan 74^\circ} \\ &= 5.73 \text{ m}\end{aligned}$$

$$CD = 25.4 - 5.73 = 19.7 \text{ m (1 d.p.)}$$

18. Measurement, STD2 M6 2012 HSC 29c

i. Find EF :

$$\begin{aligned}\angle FGE &= 180 - (139 + 31) \text{ (angle sum of } \triangle EFG) \\ &= 180 - 170 \\ &= 10^\circ\end{aligned}$$

Using Sine rule:

$$\begin{aligned}\frac{EF}{\sin 10^\circ} &= \frac{82}{\sin 139^\circ} \\ EF &= \frac{82 \times \sin 10^\circ}{\sin 139^\circ} \\ &= 21.70406\dots \\ &= 22 \text{ km (nearest km)}\end{aligned}$$

ii. Let d = total distance cycled

Find EH
Since $\triangle EGH$ is isosceles, and $\angle EHG = 90^\circ$
 $\angle GEH = \angle HGE = 45^\circ$
(angles opposite equal sides in $\triangle EGH$)

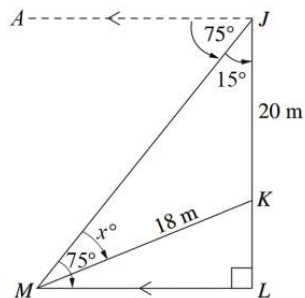
$$\begin{aligned}\sin 45^\circ &= \frac{GH}{82} \\ GH &= 82 \times \sin 45^\circ \\ &= 57.983\dots\end{aligned}$$

$$\begin{aligned}\therefore d &= EF + FG + GH + EH \\ &= 21.704\dots + 64 + 57.983 + 57.983 \\ &= 201.66\dots \\ &= 202 \text{ km (nearest km)}\end{aligned}$$

◆ Mean mark 48%.

◆ Mean mark 37%.
MARKER'S COMMENT: Students could also have used Pythagoras or the Sine rule to calculate GH

19. Measurement, STD2 M6 2007 HSC 25b



$$\angle AJL = 90^\circ$$

$$\angle MJL = 90 - 75 = 15^\circ$$

Using sine rule in $\triangle MJK$

$$\text{Let } \angle JMK = x^\circ$$

$$\frac{20}{\sin x} = \frac{18}{\sin 15^\circ}$$

$$18 \sin x = 20 \times \sin 15^\circ$$

$$\sin x = \frac{20 \times \sin 15^\circ}{18} = 0.2875 \dots$$

$$x = 16.71 \dots^\circ$$

$$\angle JML = 75^\circ \text{ (alternate angles, } ML \parallel AJ)$$

$$\therefore \angle KML = 75^\circ - 16.71 \dots$$

$$= 58.287 \dots^\circ$$

$$= 58^\circ \text{ (nearest degree)}$$

$$\therefore \text{Angle of Elevation from } M \text{ to } K \text{ is } 58^\circ.$$

20. Measurement, STD2 M6 2010 HSC 24d

$$\begin{aligned} \text{i. } \tan \angle ADB &= \frac{126}{168} \\ \angle ADB &= 36.8698 \dots \\ &= 36.9^\circ \text{ (to 1 d.p)} \end{aligned}$$

$$\angle \text{Depression } D \text{ to } B = 90 - 36.9$$

$$= 53.1$$

$$= 53^\circ \text{ (nearest degree)}$$

ii. Find CB :

$$\angle ADC + 28 = 90$$

$$\angle ADC = 62^\circ$$

$$\tan 62^\circ = \frac{AC}{168}$$

$$AC = 168 \times \tan 62^\circ$$

$$= 315.962 \dots$$

$$CB = AC - AB$$

$$= 315.962 \dots - 126$$

$$= 189.962 \dots$$

$$= 190 \text{ m (nearest m)}$$

♦♦ Mean mark 31%

♦♦ Mean mark 31%

MARKER'S

COMMENT: Solve efficiently by using right-angled trigonometry. Many students used non-right angled trig, adding to the calculations and the difficulty.

21. Measurement, STD2 M6 2010 HSC 26d

$$\begin{aligned} \cos \angle C &= \frac{AC^2 + CB^2 - AB^2}{2 \times AC \times CB} \\ &= \frac{50^2 + 40^2 - 83^2}{2 \times 50 \times 40} \\ &= -0.69725 \dots \\ \angle C &= 134.2067 \dots^\circ \end{aligned}$$

$$\text{Using Area} = \frac{1}{2} ab \sin C:$$

$$\text{Area } \triangle ABC = \frac{1}{2} \times 50 \times 40 \times \sin 134.2067 \dots^\circ$$

$$= 716.828 \dots$$

$$= 717 \text{ m}^2 \text{ (nearest m}^2)$$

♦♦ Mean mark 32%.

TIP: The allocation of 3 marks to this question should flag the need for more than 1 step.

22. Measurement, STD2 M6 2018 HSC 30c

Find AC :

$$\sin 62^\circ = \frac{AC}{13}$$

$$AC = 13 \times \sin 62^\circ \\ = 11.478\dots$$

Using the sine rule in $\triangle ACD$:

$$\text{Area} = \frac{1}{2} \times AC \times AD \times \sin 40^\circ$$

$$30 = \frac{1}{2} \times 11.478\dots \times x \times \sin 40^\circ$$

$$\therefore x = \frac{30 \times 2}{11.478\dots \times \sin 40^\circ} \\ = 8.13\dots \\ = 8.1 \text{ cm (1 d.p.)}$$

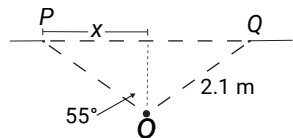
♦ Mean mark 39%.

23. Measurement, STD2 M1 2023 HSC 33

$$\text{Arc } PQ = \frac{110}{360} \times \pi \times 2.1^2 \\ = 4.03171\dots \text{ m}$$

♦ Mean mark 42%.

Consider $\triangle OPQ$:



$$\sin 55^\circ = \frac{x}{2.1} \\ x = 2.1 \times \sin 55^\circ \\ = 1.7202\dots$$

$$PQ = 2x = 3.440 \text{ m}$$

$$\therefore \text{Perimeter} = 8 + (2 \times 3.6) + 4.031 + (8 - 3.440) \\ = 23.79\dots \\ = 23.8 \text{ m (to 1 d.p.)}$$

24. Measurement, STD2 M6 2021 HSC 37

Using the sine rule:

$$\frac{\sin \theta}{25} = \frac{\sin 28^\circ}{16}$$

$$\sin \theta = \frac{25 \times \sin 28^\circ}{16}$$

$$\sin \theta = 0.73355 \\ \theta = 47^\circ$$

$$\therefore \angle ABC = 180 - 47 \\ = 133^\circ$$

♦♦ Mean mark 31%.

25. Measurement, STD2 M6 2022 HSC 26

In $\triangle ABC$:

$$\cos 41^\circ = \frac{35}{BC}$$

$$BC = \frac{35}{\cos 41^\circ} \\ = 46.375\dots$$

$$\angle BCD = 180 - 41 = 139^\circ$$

Using sine rule in $\triangle BCD$:

$$\frac{\sin \theta}{46.375} = \frac{\sin 139^\circ}{93}$$

$$\sin \theta = \frac{\sin 139^\circ \times 46.375}{93}$$

$$\therefore \theta = \sin^{-1} \left(\frac{\sin 139^\circ \times 46.375}{93} \right)$$

$$= 19.09\dots$$

$$= 19^\circ 6' \text{ (nearest minute)}$$

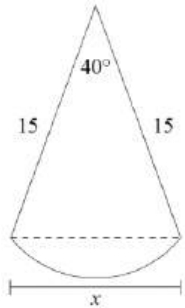
♦ Mean mark 50%.

26. Measurement, STD2 M1 2008 HSC 25c

- i. Box height = 15 cm
(radius of the arc)

$$\begin{aligned}\text{Box width} &= 3 \times 7 \\ &= 21 \text{ cm}\end{aligned}$$

$$\text{Box length} = 4x$$



Using cosine rule

$$\begin{aligned}c^2 &= a^2 + b^2 - 2ab \cos C \\ x^2 &= 15^2 + 15^2 - 2 \times 15 \times 15 \times \cos 40^\circ \\ &= 450 - 344.7199 \dots \\ &= 105.2800 \dots \\ x &= 10.2606 \dots\end{aligned}$$

$$\begin{aligned}\text{Box length} &= 4 \times 10.2606 \dots \\ &= 41.04 \dots\end{aligned}$$

$$\therefore \text{Dimensions are } 41 \text{ cm} \times 21 \text{ cm} \times 15 \text{ cm}$$

- ii. Volume = Ah

$$h = 7 \text{ cm}$$

Find A :

$$\begin{aligned}A &= \frac{1}{2} ab \sin C \\ &= \frac{1}{2} \times 15 \times 15 \times \sin 40^\circ \\ &= 72.3136 \dots\end{aligned}$$

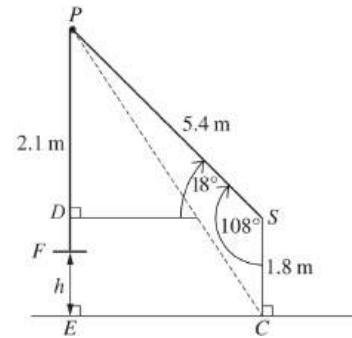
$$\begin{aligned}\therefore V &= 72.3136 \dots \times 7 \\ &= 506.195 \dots \\ &= 506 \text{ cm}^3 \text{ (nearest whole)}\end{aligned}$$

◆ Mean mark 45%.

◆◆ Mean mark 22%.

27. Measurement, STD2 M6 2015 HSC 30e

- i.



Show $PC = 6.197 \text{ m}$

Using the cosine rule in $\triangle PSC$

$$\begin{aligned}PC^2 &= PS^2 + SC^2 - 2 \times PS \times SC \times \cos 108^\circ \\ &= 5.4^2 + 1.8^2 - 2 \times 5.4 \times 1.8 \times \cos 108^\circ \\ &= 38.4072 \dots \\ \therefore PC &= 6.19736 \dots \\ &= 6.197 \text{ m (to 3 d.p.)} \dots \text{as required}\end{aligned}$$

- ii. Let $SD \perp PE$

$\angle DSC$ is a right angle

$$\therefore \angle DSP = 108^\circ - 90^\circ = 18^\circ$$

In $\triangle PDS$

$$\begin{aligned}\cos 18^\circ &= \frac{DS}{5.4} \\ DS &= 5.4 \times \cos 18^\circ \\ &= 5.1357 \dots \text{ m}\end{aligned}$$

$$EC = DS = 5.1357 \dots \text{ m (opposite sides of rectangle } DECS)$$

Using Pythagoras in $\triangle PEC$:

$$\begin{aligned}PE^2 + EC^2 &= PC^2 \\ PE^2 + 5.1357^2 &= 6.197^2 \\ PE^2 &= 12.027 \dots \\ PE &= 3.468 \dots \text{ m}\end{aligned}$$

From the diagram,

$$\begin{aligned}h &= PE - PF \\ &= 3.468 \dots - 2.1 \\ &= 1.368 \dots\end{aligned}$$

◆ Mean mark 41%.

◆◆ Mean mark below 19%.
STRATEGY: Finding PC in part (i) and needing PE to find h should flag the strategy of finding EC and using Pythagoras.

= **1.37 m** (to 2 d.p.)

